GHANA PLASMODIUM INTELLIGENCE PLATFORM

P-TRANSMIT AI

The AI Model — Information Document

Purpose, Development & Technical Overview

Document Type:	AI Model Information & Technical Overview
Platform Version:	0.1.0 — Phase-1 MVP
Date:	28 October 2026
Model Family:	Supervised Tabular Regression
Algorithms:	Random Forest, XGBoost, Ridge Regression
Target Variable:	Weekly confirmed malaria cases per region
Forecast Horizons:	1, 2, 4, and 8 weeks ahead
Coverage:	Ghana — 16 administrative regions (ADM1)
Validation Method:	Strictly chronological (temporal) hold-out
Code Location:	backend/app/ml/transmission.py
License:	MIT (code) · geoBoundaries CC-BY 4.0 (boundaries)

1. Overview

1.1 What is P-TRANSMIT AI?

P-TRANSMIT AI is an open, locally-hostable decision-support platform for Plasmodium falciparum transmission, genomic epidemiology, and antimalarial drug resistance surveillance in Ghana. It is designed for public-health researchers and officers who need forward-looking, region-level insights from weekly surveillance data.

The platform ingests weekly surveillance data (confirmed cases plus optional environmental covariates such as rainfall, temperature, humidity, and vegetation index), trains and compares supervised forecasting models, produces per-region forecasts with uncertainty ranges, explains what drove each prediction, and raises relative early-warning signals for human review.

1.2 Platform Modules

Dashboard
National KPIs, choropleth map, national forecast, per-region explanations, alerts overview.

Transmission AI
Trains models, shows held-out observed-vs-predicted curves and metrics per horizon.

Spatial Intelligence
Interactive map of Ghana's 16 regions with risk / observed / forecast layers.

Forecasting
1/2/4/8-week forecasts with residual-based uncertainty intervals and stated assumptions.

Explainable AI
SHAP (when available) or perturbation-based per-region contributions — model behaviour, not causality.

Early-warning Alerts
Level / trend / forecast signals relative to the dataset's own history; human review required.

Data Explorer
CSV/TSV/XLSX/JSON upload, data-quality report, column mapping, Ghana region fuzzy-matching.

Model Laboratory
Train/compare/manage models with full reproducibility provenance.

Reports
Markdown research report including a limitations section.

Genomics / Resistance
Scaffold modules awaiting validated data (no simulated results).

2. The AI Model

2.1 Model Architecture — Ten Layers

The 'AI model' is not a single neural network. It is a pipeline of ten layers, each with a specific job. Together they transform raw weekly surveillance rows into forecasts, explanations, and early-warning signals.

Layer 1 — Data

Standardised weekly rows: one row per region per epi-week. Required columns: region, week_start_date, malaria_cases. Optional covariates include rainfall, temperature, humidity, NDVI, population, suspected cases, and test positivity rate.

Layer 2 — Features

For every region-week, the engine constructs: case history lags (1-4 weeks), current + lagged environmental conditions, cyclic seasonality encoding (sin/cos of week-of-year), and region fixed-effect dummies. The target is cases at week t+h (the forecast horizon).

Layer 3 — Models

Three competing algorithms train per horizon: Ridge (regularised linear baseline), Random Forest (250 trees, bagged ensemble), and XGBoost (500 boosted trees). The best model per horizon is selected by lowest test MAE.

Layer 4 — Validation

Strictly chronological split: earliest 80% of weeks for training, latest 20% for testing. Train and test weeks are disjoint — no future information can leak backward. This is verified by automated tests.

Layer 5 — Metrics

MAE and RMSE are primary metrics for case counts. MAPE, sMAPE, and R-squared are also reported. R-squared is explicitly noted as 'for completeness only' — low R-squared is normal for noisy count data.

Layer 6 — Forecast

At the last observed week, the engine builds one feature row per region and predicts cases h weeks ahead. Environmental features are held at their last observed value (persistence assumption), which is stated in every forecast response.

Layer 7 — Uncertainty

Each forecast carries interval_low and interval_high = prediction +/- 1.96 * residual_std. These are residual-based Gaussian approximations from held-out test errors — NOT calibrated probabilistic intervals.

Layer 8 — Risk

Risk categories are RELATIVE to the loaded dataset's own historical distribution. Predicted >= 90th percentile = 'very high'; >= 75th = 'high'; >= 50th = 'moderate'; else 'low'. These are not national clinical thresholds.

Layer 9 — Explainability

Per-region explanations show top 8 features driving each prediction. Uses SHAP TreeExplainer when available (tree models), otherwise single-feature perturbation. Positive = pushed prediction up; negative = pulled down. Describes model behaviour, NOT causal effects.

Layer 10 — Alerts

Three deterministic rules scan every region (minimum 30 weeks history): Level alert (last 4 weeks mean > historical p90), Trend alert (8-week slope > 90th percentile of historical slopes), and Forecast alert (median 1-4 week prediction > p90). All require human review.

3. Machine Learning Algorithms

3.1 Ridge Regression (Baseline)

A regularised linear model using StandardScaler + Ridge (alpha=1.0). Learns one weight per feature, making it fully interpretable. It captures linear relationships only but serves as an honest, stable baseline. Training is fast and the model rarely overfits. If non-linear methods do not substantially beat Ridge, it suggests the predictive signal is predominantly linear.

3.2 Random Forest

An ensemble of 250 decision trees using bagging (bootstrap aggregating). Key parameters: min_samples_leaf=2, random_state fixed for reproducibility, n_jobs=-1 for parallel training. Random Forest captures non-linear relationships and interactions (e.g., rainfall × season × region effects) and is robust to outliers. It is less interpretable than Ridge but more stable than boosting on small datasets.

3.3 XGBoost (Gradient Boosting)

A gradient boosting machine with 500 trees, learning rate 0.05, max depth 4, subsample 0.9, colsample_bytree 0.9, and L2 regularization (lambda=1.0). Uses histogram-based tree construction for speed. XGBoost usually performs best on tabular data but is the most sensitive to hyperparameter settings and data quality. Trees are added sequentially, each correcting the previous tree's errors.

3.4 Model Selection

One model is trained per algorithm per horizon, producing up to 12 model fits per training run (3 algorithms × 4 horizons). For each horizon, the model with the lowest test-set MAE is selected as the 'best' and persisted as a joblib artifact with full provenance metadata. A fixed random seed (default 42, user-settable) ensures reproducibility across runs.

4. Feature Engineering

4.1 Feature Groups

The model does not see raw data directly. For each region-week, the following features are constructed:

Case History (Lags)
cases_lag1 through cases_lag4 — confirmed malaria cases in each of the previous 1-4 weeks for the same region. These are typically the strongest predictors because recent transmission momentum is highly predictive of near-future cases.

Environment (Current)
rainfall_mm, temperature_mean_c, humidity_pct, ndvi — current-week environmental conditions. These are known correlates of malaria transmission dynamics.

Environment (Lagged)
rainfall_lag1, temperature_lag1, humidity_lag1, ndvi_lag1 — same environmental variables from one week prior, capturing delayed effects.

Seasonality
sin_week and cos_week — cyclic encoding of ISO week-of-year using sine and cosine transformations with period 52. This allows the model to learn seasonal patterns without discontinuities at year boundaries.

Region Fixed Effects
One-hot encoded dummy variables (region_Ashanti, region_Brong_Ahafo, etc.) — 16 binary indicators, one per Ghana region. This allows a single global model to capture each region's baseline transmission level.

Population (Optional)
population_illustrative — when present, used as a denominator or weighting factor.

4.2 Target Construction

For forecast horizon h (where h is 1, 2, 4, or 8 weeks), the target variable is constructed by shifting the malaria_cases column backward by h weeks (shift(-h)). This means: at week t, the model learns to predict cases at week t+h. Rows where the target cannot be computed (because h weeks of future data don't exist) are excluded from training.

4.3 Data Requirements

The minimum viable dataset needs: (1) region names matching Ghana's 16 administrative regions, (2) week_start_date in ISO date format, and (3) malaria_cases as integer counts. The Data Explorer auto-detects column synonyms and fuzzy-matches region names. A minimum of 30 weeks of history per region is needed for meaningful alert generation, and the first 4+ weeks per region are consumed by lag construction.

5. Validation & Metrics

5.1 Temporal Validation Strategy

The platform uses a strictly chronological train-test split — never random. The earliest 80% of unique weeks (configurable as 'train fraction') form the training set; the latest 20% form the test set. Train weeks and test weeks are guaranteed to be disjoint, and the maximum training week is always less than the minimum test week. This is verified by automated tests that assert no data leakage occurs.

This approach approximates how the model would perform in production: it learns from the past and is evaluated on the future. Random shuffling of time-series data is never performed, as it would allow the model to 'see' future patterns during training.

5.2 Performance Metrics

Metric	Formula / Definition	How to Interpret
MAE	Mean Absolute Error: avg|pred-obs|	Typical error in cases/week. Compare to region's case scale.
RMSE	Root Mean Squared Error: sqrt(avg(e²))	Like MAE but penalises large misses. Large gap vs MAE = occasio
MAPE	Mean Abs Pct Error (non-zero weeks)	Percentage error. Unstable near zero; excluded on zero weeks.
sMAPE	Symmetric Mean Abs Pct Error	Bounded percentage variant for cross-region comparison.
R^2	Coefficient of determination	For completeness only. Low R^2 normal for noisy count data.
Residual Std	Std dev of test-set residuals	Sizes uncertainty intervals. Larger = wider forecast bands.

6. Forecasting System

6.1 How Forecasts Are Produced

Once models are trained and persisted as joblib artifacts, the forecasting system works as follows:

Step 1:
At the last observed week (t), the engine constructs one feature row per region using actual historical values for case lags and the most recent environmental data.

Step 2:
Seasonality features (sin_week, cos_week) are computed from the calendar for the forecast week.

Step 3:
Each persisted best model (one per horizon: 1, 2, 4, 8 weeks) predicts the case count for its respective horizon.

Step 4:
Environmental features are held at their last observed value — this is the 'persistence assumption' and is explicitly stated in every forecast response.

Step 5:
Predictions are clipped at zero (case counts cannot be negative).

Step 6:
Uncertainty intervals are computed as prediction +/- 1.96 × residual_std, where residual_std comes from the model's test-set errors.

6.2 Environmental Persistence Assumption

A critical limitation: the forecasting system does not know future weather. When producing a forecast for week t+h, all environmental covariates (rainfall, temperature, humidity, NDVI) are held constant at their last observed value from week t. This means: genuine weather shocks (unseasonal heavy rain, heat waves, etc.) will NOT be anticipated by the model. The assumptions are clearly stated in every API response and dashboard display.

6.3 Uncertainty Intervals

Each forecast row includes interval_low and interval_high computed as: prediction +/- 1.96 × residual_std (with the lower bound clipped at zero). The residual_std is the standard deviation of the model's errors on held-out test weeks — i.e., 'how wrong this model has actually been on unseen data.'

These intervals are explicitly labelled as 'residual-based Gaussian approximation (not calibrated probabilistic intervals).' They show the model's typical error range but are NOT 95% probability guarantees. Treat values near the upper bound as 'worth watching' rather than as certain outcomes.

7. Explainable AI

7.1 Global Feature Importance

The platform computes permutation importance on held-out test weeks (5 repeats, scored by MAE change) to rank which features the model relies on overall. The top 12 features are displayed. Permutation importance is model-agnostic: it measures how much prediction accuracy drops when a feature's values are randomly shuffled, breaking its relationship with the target.

7.2 Per-Region Explanations

For each region, the dashboard shows the top 8 features driving that region's current prediction as signed bars. The method works as follows:

1. A base prediction is computed for the region using all feature values.
2. For each non-region feature, that feature is replaced with its training-set median value.
3. The model re-predicts with the replaced feature.
4. The difference from the base prediction is recorded as that feature's contribution.
5. Contributions are sorted by absolute magnitude; top 8 are displayed.
6. Positive contributions pushed the prediction UP; negative contributions pulled it DOWN.

7.3 SHAP Integration (Optional)

When the optional 'shap' Python package is installed, tree-based models (Random Forest and XGBoost) use SHAP TreeExplainer instead of the perturbation heuristic. SHAP (SHapley Additive exPlanations) provides game-theory-based feature attributions with stronger theoretical foundations. The method used is always displayed alongside the explanations.

8. Early-Warning Alert System

8.1 Alert Rules

Three deterministic rules scan every region with at least 30 weeks of history. All rules are relative to the loaded dataset's own historical distribution — no absolute thresholds are invented.

Rule	Severity	Trigger Condition
Level	Warning	Mean of the last 4 observed weeks EXCEEDS the region's historical 90th percentile of weekly case
Trend	Warning	Least-squares slope of the last 8 weeks EXCEEDS the 90th percentile of all historical 8-week slope
Forecast	Info	Median model prediction for weeks 1-4 EXCEEDS the region's historical 90th percentile. Flags whe

8.2 Alert Workflow

1. Alerts are generated by running an 'alert scan' which applies all three rules to every region.
2. Each alert stores: date, region, severity, alert type, evidence numbers (the specific values that fired the rule), model reference, and a plain-language explanation.
3. Duplicate suppression: if an open alert of the same type and region already exists, a new scan will NOT create a duplicate.
4. Workflow states: Open → Acknowledged (kept on radar) or Dismissed (not actionable). A human always makes this decision.
5. Alerts are research signals, NOT confirmed outbreaks. They require human and public-health review before any action.

9. Scientific Governance

9.1 Principles Enforced in Code

- Synthetic demonstration data are always labelled clearly and never mixed with real data. The demo dataset name includes 'DEMONSTRATION DATA — NOT REAL SURVEILLANCE DATA.'
- No invented thresholds. Risk categories are relative to the loaded dataset's own history. Resistance interpretation requires researcher-supplied definitions — the platform never assigns clinical meaning to genetic markers.
- Strictly temporal validation. Time-series rows are never shuffled. Train weeks always precede test weeks. This is verified by automated tests.
- MAE and RMSE are primary metrics for count outcomes. R-squared is reported for completeness only and explicitly noted as such.
- Explanations describe model behaviour, not causal effects. The method text always ends with this clarification.
- Alerts are research signals, not confirmed outbreaks. They require human and public-health review. No automatic actions are taken based on alerts.
- Genetic marker does not equal phenotypic drug resistance. The genomics and resistance modules are descriptive scaffolds awaiting validated data.
- Audit logs record logins, dataset operations, training runs, and report generation for traceability.

9.2 Data Provenance & Reproducibility

Every trained model artifact (saved as a .joblib file) contains complete provenance metadata: the exact feature list used, train/test week ranges, random seed, software versions (Python, scikit-learn, XGBoost, pandas, numpy), and UTC training timestamp. This makes every training run fully reproducible — running the same data with the same seed produces identical results.

Artifact files follow the naming convention: transmission_{model_type}_h{horizon}_ds{dataset_id}.joblib. Each contains the fitted model, model type, horizon, feature names, all metrics, feature importance rankings, train/test week lists, seed, software versions, and training timestamp.

10. Technical Architecture

10.1 Technology Stack

Backend / API:	Python 3.12, FastAPI, SQLAlchemy (SQLite default, PostgreSQL optional), scikit-learn, XGBoost, pandas, numpy, joblib
Frontend:	React 18, TypeScript, Vite, Recharts (charts), Leaflet (interactive maps)
Geography:	geoBoundaries gbOpen GHA ADM1 boundaries (16 regions), CC-BY 4.0 license, simplified for web map rendering
Model Artifacts:	joblib format containing fitted estimators + full provenance metadata
Deployment:	Single Docker image serving both API (/api/*) and built React UI (/) from one port. Alternatively, separate backend + frontend for development.

10.2 API Endpoints

GET /api/health	Health check — returns status, app name, version, demo mode flag, and disclaimer.
GET /api/datasets	List all uploaded datasets with metadata and data-quality reports.
POST /api/datasets/upload	Upload CSV/XLSX/JSON surveillance or genomic data with auto-detection of columns and region matching.
GET /api/models	List trained models with metrics and provenance.
POST /api/models/train	Train transmission models on a dataset — returns results per horizon per model.
GET /api/forecasts	Retrieve forecasts for selected horizons using persisted best models.
GET /api/explain	Get per-region feature contributions for the current forecast.
GET /api/alerts	List all alerts with status, severity, evidence, and explanations.
POST /api/alerts/scan	Run alert scan — applies all rules to current data and creates new alerts.
GET /api/reports/generate	Generate a Markdown research report with dataset info, models, alerts, and limitations.

11. Development & Roadmap

11.1 Current State (Phase 1 MVP)

The current version (0.1.0) is a Phase-1 Minimum Viable Product focused on malaria transmission forecasting for Ghana's 16 regions. It provides a complete pipeline from data upload through model training, forecasting, explainability, and alerting — all running locally or on free hosting tiers.

The genomics and drug resistance modules exist as scaffolds: they accept variant table uploads and compute descriptive statistics (marker counts, frequencies per stratum) but do NOT perform resistance interpretation. This is intentional governance — no resistance claims are made without researcher-supplied validated definitions.

11.2 Known Limitations

1. Environmental persistence: forecasts hold weather/vegetation at last observed values; genuine rainfall shocks will not be anticipated.
2. Direct (non-recursive) horizons: each horizon is a separate model; an 8-week forecast does not internally compound the 1-4 week forecasts.
3. Intervals are uncalibrated Gaussian approximations from test residuals, not true probabilistic forecasts.
4. Regional dummy coverage: a region absent at training time gets no dedicated dummy — all 16 regions should have history before real use.
5. Minimum history requirements: alert rules need ≥ 30 weeks; lag construction consumes the first 4+ weeks per region.
6. Demo data are synthetic with a scripted late-season upswing to exercise alerts — they are always labelled and must never be used for research conclusions.
7. Genomics/Resistance modules are descriptive scaffolds awaiting validated data; detected markers are not phenotypic resistance.
8. Free-tier hosting has ephemeral storage — trained models and uploaded accounts do not survive restarts (local storage in models/ persists).

11.3 Future Roadmap

Phase 2

Genomic module build-out: SNP statistics, diversity indices (H, MOI), population structure analysis; resistance surveillance with researcher-defined thresholds.

Phase 3

Environmental raster ingestion (CHIRPS rainfall, ERA5-Land temperature, MODIS NDVI), Bayesian spatial models, spatiotemporal deep learning models, calibrated uncertainty quantification, configurable alert thresholds.

Phase 4

Multimodal AI combining genomic + clinical + environmental data, mechanistic transmission model coupling, PDF report export, multi-user deployment hardening with role-based access control.

12. Running the Platform

12.1 Local Execution (Windows)

Double-click start-ptransmit.bat in the project root. This launches both the backend (FastAPI on port 8000) and the frontend (Vite preview on port 4173) and opens the browser to http://localhost:4173. Keep both server windows open — close them to stop the platform.

12.2 Single-Process Mode (Backend Serves UI)

After building the frontend (cd frontend && npm run build), start only the backend: cd backend && python -m uvicorn app.main:app --host 0.0.0.0 --port 8000. The backend serves both the API and the built React UI from one address: http://127.0.0.1:8000/. This is the recommended mode for hosting and the mode currently running.

12.3 Current Access Points

This platform instance is currently running with the following access points:

Full Platform (Single Link)

URL: http://127.0.0.1:8000/

Backend serves both API and built React UI — ONE link opens everything.

Frontend Only (Dev Preview)

URL: http://127.0.0.1:4173/

Vite preview server; proxies API calls to port 8000.

API Health Check

URL: http://127.0.0.1:8000/api/health

Returns status, version, demo mode flag, and disclaimer.

API Documentation

URL: http://127.0.0.1:8000/docs

Interactive Swagger UI for all API endpoints.

12.4 Default Credentials

Default accounts (change immediately for any real deployment): Admin: admin / ptransmit-admin | Researcher: demo / demo1234. In demo mode, you can also click 'Enter live demo' for read-only access without an account.

13. Data Specifications

13.1 Input Data Requirements

Upload a weekly surveillance table (CSV, XLSX, JSON, or TSV) with at minimum: region (Ghana region, fuzzy-matched to 16 canonical names), week_start_date (ISO date), and malaria_cases (confirmed case count). Synonyms are auto-detected for all columns.

13.2 Column Specifications

Standard Column	Purpose	Recognised Synonyms
region	Ghana region name	region_name, admin1, adm1, province
week_start_date	ISO date of epi-week start	date, week_start, report_date
malaria_cases	Confirmed cases (count)	mal_cases, cases, confirmed_cases
suspected_cases	Tests conducted / suspected	tested, suspected
test_positivity_rate	Test positivity percentage	positivity
rainfall_mm	Rainfall (mm)	rainfall
temperature_mean_c	Mean temperature (°C)	temp, temperature
humidity_pct	Relative humidity (%)	rh, humidity
ndvi	Normalised Difference Vegetation Index	evi, vegetation_index
population_illustrative	Denominator / population weight	population, pop

14. Summary

P-TRANSMIT AI is a thoughtfully engineered decision-support platform that brings together machine learning, explainable AI, and public-health domain knowledge to address malaria surveillance in Ghana. Its key strengths are:

- A complete, locally-runnable pipeline from data upload to actionable insights — no cloud dependency required.
- Scientific honesty: temporal validation, relative risk categories, clearly stated assumptions, and explicit disclaimers about what predictions are and are not.
- Reproducibility: every model training run records its full provenance — data, split, seed, software versions, and timestamp.
- Explainability: per-region explanations show what drove each prediction, with the method always disclosed (SHAP when available, perturbation fallback otherwise).
- Governance by design: synthetic data is always labelled, no invented thresholds, alerts require human review, and genetic markers are never equated with drug resistance.
- Open source (MIT license) with clear data credits (geoBoundaries CC-BY 4.0, OpenStreetMap tiles).

This document was generated from backend/app/ml/transmission.py and backend/app/services/alerts.py on 28 October 2026. Code: MIT (c) 2026 P-TRANSMIT AI contributors. Boundaries: geoBoundaries gbOpen GHA ADM1, CC-BY 4.0. Research and surveillance decision support — not a diagnostic device.